Problem with Money
(a) Segregation

- **Banking Access Disparity**: Around 38% of the world's population lacks bank accounts, leading to limited access to banking services like online payments and loans.
- **Economic Impact**: Without bank accounts, people can't expand businesses online or access wider markets, limiting economic growth.
- **Daily Challenges**: Lack of banking services forces people in rural areas to travel long distances for basic transactions like paying bills or topping up mobile phones.
- **Cost and Documentation Barriers**: Banking services are often too expensive, and many people lack the necessary documentation, such as ID cards, to open accounts.
- **Discrimination in Banking**: People can be denied banking services due to political views, business types, ethnicity, nationality, sexual orientation, or skin color.

Problem with Money
(b) Privacy

- **Privacy Concerns**: Traditional electronic payments can be easily traced, censored, frozen, or seized by the state, posing significant privacy issues.

- **Government Power**: The ability of governments to control funds can be abused, especially with changing laws and political climates.

- **Examples of Abuse**: Notable cases include WikiLeaks' 2010 donation blockade by major payment networks under US government pressure and Cyprus seizing 47.5% of large bank deposits in 2013.

- **Cash Transactions**: Cash remains a private and free means of trade, though some regions, like Sweden, are phasing out cash, leading to complete transaction traceability.

- **Future Risks**: The potential for government overreach in financial controls emphasizes the need for privacy in electronic payments.

Problem with Money
(c) Borders

- **Cross-Border Transfer Challenges**: Moving fiat currency across national borders is difficult, expensive, and sometimes prohibited.

- **High Transfer Fees**: Using services like Western Union to send money internationally incurs high fees, ranging from 4.9% to 16.3%, especially if the recipient can only receive cash.

- **Fee Comparison**: Sending 1,000 SEK from Sweden to the Philippines can cost significantly more depending on the transfer method and recipient's access to banking services.

- **Domestic Transfers**: Within a country's borders, transferring fiat currency is generally easy and convenient, often facilitated by direct cash handovers or mobile apps.

- **Currency Efficiency**: Fiat currencies are effective for domestic transactions, maintaining ease of use and low cost as long as transfers remain within the same country and currency.

Approach of Bitcoin
(a) Decentralized

- **Decentralized Control**: Bitcoin is controlled by thousands of nodes with equal privileges, making it decentralized unlike centralized systems like banks or the Federal Reserve.

- **Equality and Accessibility**: In a decentralized system like Bitcoin, all users are treated equally regardless of location or identity, and it's hard to control or restrict usage.

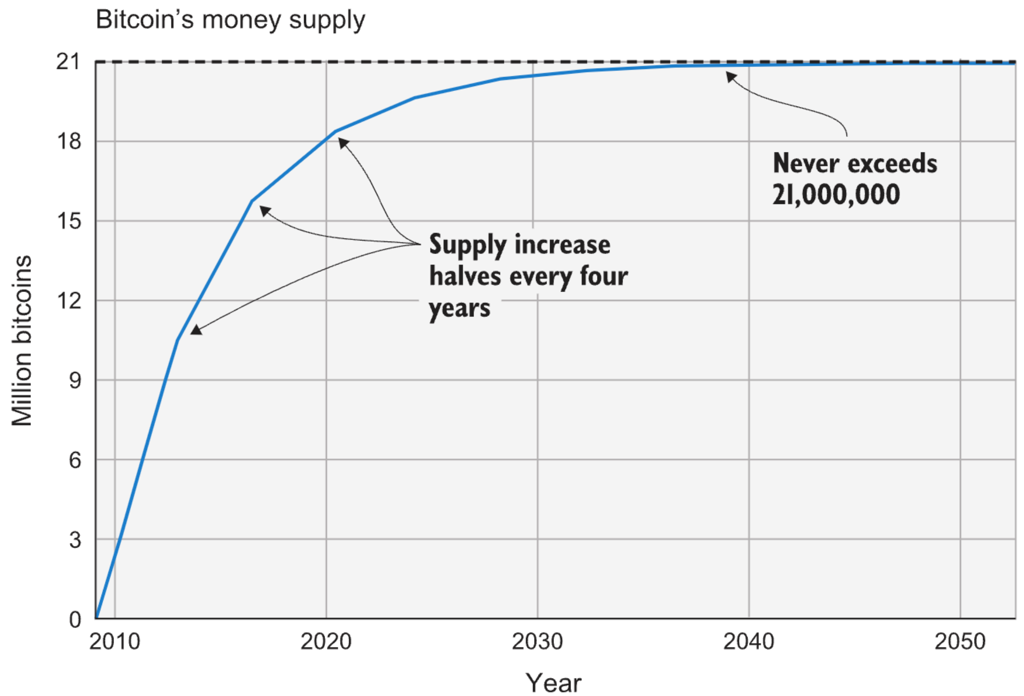
- **Permissionless Participation**: Anyone with a computer and internet connection can set up a Bitcoin node and join the network without needing permission or registration.

- **Resistance to Censorship**: Bitcoin’s decentralized nature means there’s no central point to censor payments, deny service, or seize funds.

- **Rule Stability**: Changing Bitcoin’s rules, such as its 21 million coin limit, is nearly impossible without broad consensus, ensuring stability and resistance to manipulation.

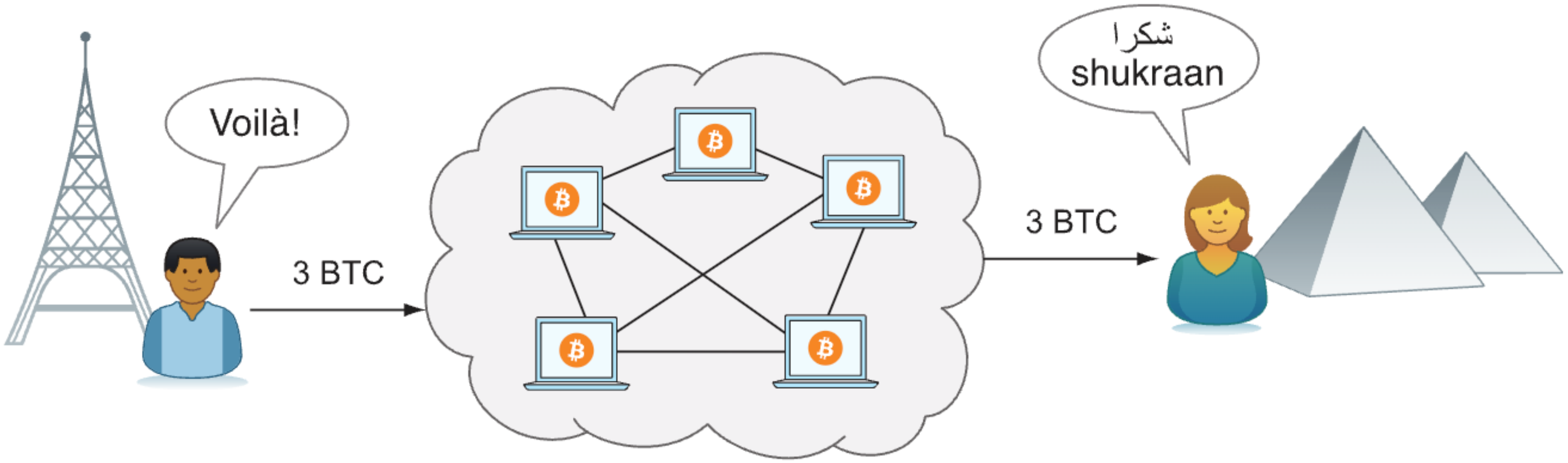
Approach of Bitcoin
(a) Decentralized

- **Limited Supply**: Bitcoin's total supply is capped at 21 million, ensuring each bitcoin represents a fixed portion of the total supply.
- **Inflation Resistance**: Unlike fiat currencies, Bitcoin's supply cannot be arbitrarily increased, making it resistant to high inflation.
- **Scheduled Growth**: Bitcoin's supply is currently growing at a diminishing rate, set to stop increasing around the year 2140.
- **Current Supply**: As of now, there are about 17 million bitcoins in circulation, with the supply increasing by approximately 4% annually, halving every four years.



Approach of Bitcoin
(b) Borderless

- **Global Reach**: Bitcoin operates globally, allowing anyone with an internet connection to send money anywhere in the world.
- **Uniform Experience**: Sending bitcoin is the same whether to someone in the same room or on another continent, with near-instantaneous payment visibility.
- **Quick Confirmation**: Recipients can be sure the money is theirs within about 60 minutes.
- **Irreversible Transactions**: Once settled, Bitcoin transfers cannot be reversed without the recipient's consent.



Bitcoin Usage
(a) Saving

- **Private Key Storage**: Bitcoin security relies on storing private keys, which can be written on paper, stored electronically, or memorized.
- **Savings Use Case**: Bitcoin is ideal for savings, with a simple method being writing down a private key on paper and storing it securely.
- **Security and Convenience**: Various saving schemes balance security and convenience, from unencrypted mobile storage to encrypted keys in a secure vault.
- **Key Safety**: The safety of your Bitcoin depends entirely on keeping your private keys secure.

Bitcoin Usage
(b) Borderless Transfer

- **Costly Traditional Transfers**: Moving money internationally, especially to poor countries without bank accounts, is expensive, often around 15%.

- **Bitcoin for Remittances**: Using Bitcoin to transfer money is becoming popular as it is usually cheaper and faster than traditional methods.

- **Service Providers**: Some companies facilitate transfers by converting currencies to Bitcoin and back, charging lower fees than traditional services.

- **Direct Transfers**: Bitcoin allows direct transfers without intermediaries, reducing costs and embodying the essence of decentralized finance.

Bitcoin Usage
(c) Shopping

- **Ideal for Shopping**: Bitcoin's borderlessness and security make it perfect for online payments for goods and services.

- **Traditional Payments Risk**: Debit card purchases involve sharing card details with merchants, increasing the risk of data breaches.

- **Data Storage Vulnerability**: Merchants store card details in databases, which can be hacked, leading to stolen card information.

- **Bitcoin Security**: Bitcoin transactions don't involve sending sensitive information, reducing the risk of data theft and ensuring secure payments.

When not to use Bitcoins
(a) Small Payments

- **Transaction Fees**: Bitcoin transactions include processing fees based on the transaction size in bytes, not the amount sent.

- **Fee Factors**: The fee is influenced by supply and demand for blockchain space, with higher fees likely prioritizing transactions.

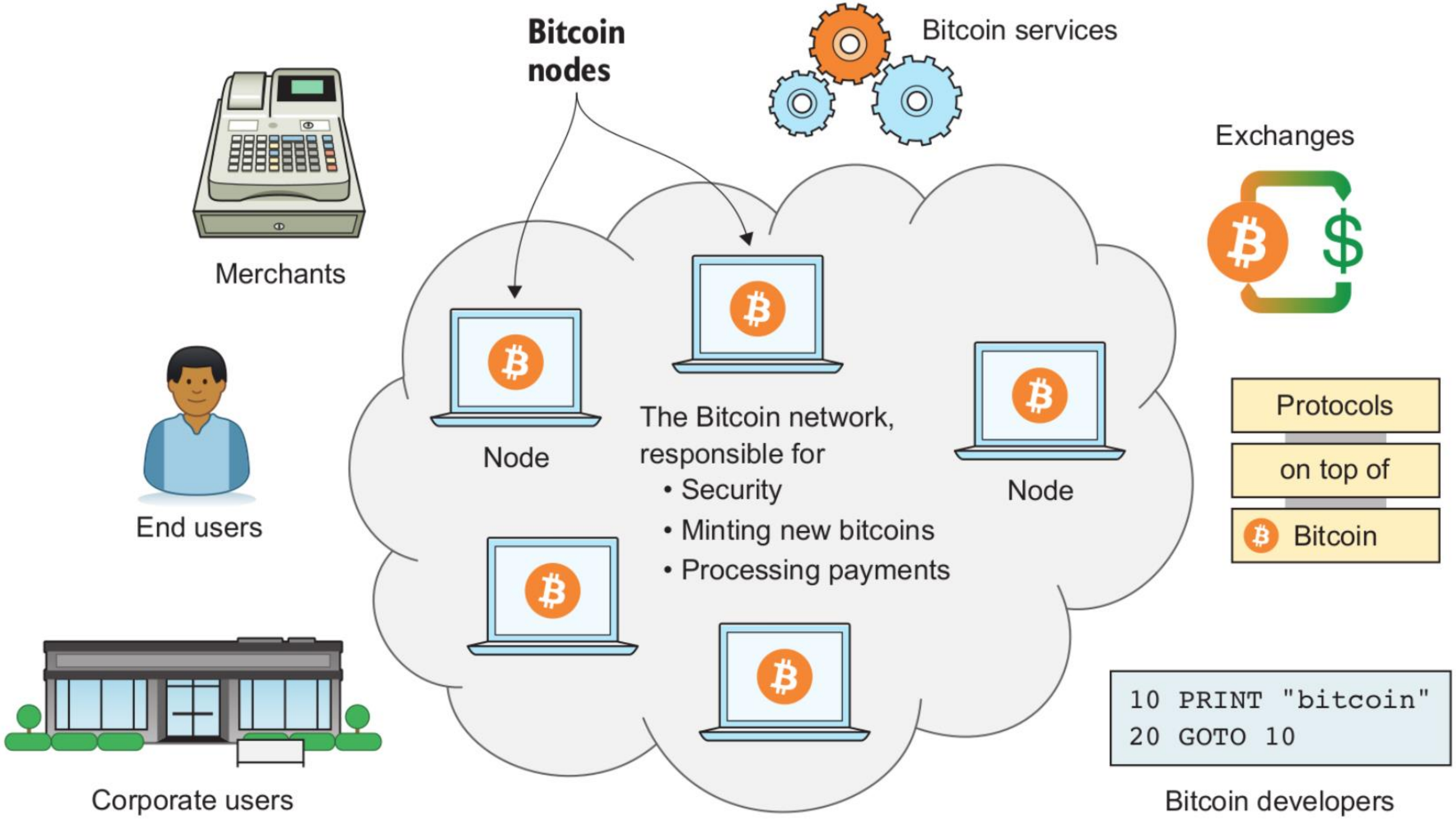
- **Economic Viability**: High fees relative to payment amount make ordinary Bitcoin transactions less viable for small payments.

- **Emerging Solutions**: Technologies like the Lightning Network enable cheap, instantaneous micropayments, significantly reducing fees for small transactions.

When not to use Bitcoins
(a) Instant Payments

- **Confirmation Time**: Bitcoin payments take about 20 minutes to confirm, during which the transaction is visible but not fully trusted.
- **Double Spending Risk**: Without confirmation, there's a risk of double spending, where the sender could use the same bitcoins in another transaction.
- **Impact on Retail**: The confirmation delay is inconvenient for brick-and-mortar shops, as customers don't want to wait for their purchases.
- **Solutions**: Technologies like the Lightning Network can address this issue, providing fast, secure transactions, especially for small payments.

Customers



Bitcoin nodes
Bitcoin services
Exchanges
Merchants
Node
The Bitcoin network, responsible for
• Security
• Minting new bitcoins
• Processing payments
Node
Protocols
on top of
Bitcoin
End users
Corporate users
10 PRINT "bitcoin"
20 GOTO 10
Bitcoin developers

- **End users:** People using Bitcoin for their day-to-day needs, such as savings, shopping, speculation, or salaries
- **Corporate users :** Companies using Bitcoin to solve their business needs, such as paying wages internationally, or use cases similar to those of end users
- **Merchants:** For example, a restaurant or a bookstore accepting Bitcoin payments
- **Bitcoin services:** Companies providing Bitcoin-related services to customers, such as topping up mobile phones, anonymization services, remittance services, or tipping services
- **Exchanges :** Commercial services people can use to exchange their local currency to and from bitcoins
- **Protocols on top:** Systems that operate "on top" of Bitcoin to perform certain tasks, such as payment network protocols, specialized tokens, and decentralized exchanges
- **Bitcoin developers:** People working, often for free, with the open source computer programs that participants of the Bitcoin network use